ReNu-Biome portal — developer handover: from the November 2 build to the approved design

Nov 2, 2026 · @radhika gundavelli

1. Purpose and how to read this

This is the one document for finishing the ReNu-Biome customer portal: it takes the build as it stood on November 2, 2026 at renu-app-1.onrender.com / renu-app-backend.onrender.com and describes, screen by screen and item by item, what has to change to reach the approved design for the February 2027 release (before almond bloom) and the Phase 2 features after it. It replaces the earlier requirements document and the November 2 build check; where they differ, this document wins.

How it is organised:

- **Section 2** is where the build is today: what is done and what is not.
- **Section 3** is the target in one page: the decisions that set the scope.
- **Sections 4 and 5** are the screens, with the prototype screenshots and, under each, what to build or change. The prototype is the visual reference for every screen; the text under each screenshot is the requirement.
- **Section 6** is the work list in the order to do it, each item with what "done" looks like.
- **Section 7** is accounts, hosting, email and integrations: what ReNu-Biome has already provided and what it still owes.
- **Section 8** is the acceptance tests the owner runs before release.
- **Section 9** is the data model and the calculation rules (products, programs, nutrient accounting).

Attached with this document (in the PDF version, both files are embedded: open the Attachments panel in Acrobat or Preview to save them):

- **grower-portal-prototype-v2.html** (October 21, 2026): the clickable prototype. Open it in any browser. Sign in with the prototype shortcuts; More → Company admin opens the admin console; the switch under More → Prototype v2 hides the Phase 2 items to show the February release alone. Everything in it is fictional sample data (Sierra Orchards LLC, Maria Lopez, two ranches).
- **renu-biome-brand-assets.zip**: logo in light and dark versions, the app icon at 1024/512/180/64 px, the install QR code (PNG and SVG), and the stock bloom photo (not licensed; a placeholder until ReNu-Biome supplies its own).

Terms: **owner** is the ReNu-Biome owner who manages Render, GitHub and this project; **office** is the owner plus one office person, the two admin users; **CEO** holds the Microsoft 365, OptimoRoute and Otodata admin logins; **customer** is a grower with one login per business. Requirement codes in brackets (A2.3, B4.2) keep the numbering of the earlier document so nothing already agreed is lost: A-items are the February release, B-items are Phase 2.

2. Where the build is today (checked November 2, 2026)

The build was checked on November 2 at renu-app-1.onrender.com and renu-app-backend.onrender.com, signed in with the staging Grower and Admin demo logins, every tab and form opened, and the API called directly for feature flags, products, programs, lab results, the dashboard and login. Nothing was changed in the build during the check. Most of the February scope exists in some form; what remains is a short list of blockers, a set of screens that do not yet match the prototype, and the Phase 2 features.

Done, and needs no further work

- Sign-in page to the prototype: bloom photo, teal, "Farmer's Pride" with GROWER PORTAL, email and password with show/hide, Forgot password, "Need an account? Email support@renu-biome.com", no Sign Up, Apple or Google buttons.
- Data model: ranch → blocks (block ID, crop, variety, rootstock, acreage, planting year, irrigation, fertigation set, county, APNs, kernel yield) and ranch → tanks; ranch acreage is the sum of its blocks; no map; no entity switcher; block detail opens from the ranch table.
- Deliveries tab with Scheduled → Out for delivery → Delivered plus Cancelled with reason, the three-stage timeline, ranch filter, upcoming then past.
- Invoices with Open / Paid / All, Overdue status, linked delivery, Year-End Summary.
- Admin console: Today (tiles with icons, deliveries due with gate code and notes, Mark delivered → delivered gallons → invoice number, dates, amount, QuickBooks ID; overdue invoices with Mark paid), Deliveries (filters, Dispatch, Mark delivered, Reschedule, Cancel with reason), Customers (contact, ranches, acres, YTD gallons, open balance; Add customer; Ranches & blocks; Create order; Reset password), Ranches (Add ranch, block, tank), Products (add, edit, delete, OMRI, photo), Settings with the feature-flag screen (global and per customer) and CSV export by date range.
- Install page at /install when signed out, with the tagline, QR code and the app.renu-biome.com/install address; web app manifest (ReNu-Biome, standalone, #0C6054, the supplied 180 and 512 icons) and a registered service worker.
- Sample submission form on a block: ranch, block, sample type (leaf tissue, soil, petiole, irrigation water), date, lab, notes, file upload.
- The four monthly reports exist as a screen (sales by customer, sales by product, receivables aging, gallons by month) with CSV export.

Not done: the short version (the full list, with what to do, is section 6)

Area	State on November 2
Security	The Grower demo login uses a literal "mock_token" that the backend accepts for admin routes; a grower can read another customer's ranch and delivery. No rate limit on login; token life 7 days; security headers not verifiable.
Feature flags	All off in the API, but the customer UI shows every flagged feature (ordering, chat, ILRP reports, SDS vault, Otodata, "OptimoRoute Live").
Invoices	No PDF upload; View PDF renders an app-generated template with a made-

Area	State on November 2
	up email. Two invoices share INV-2026-001.
Orders	Create Order from Today opens the Deliveries tab, not the form; the form has free-text customer and product, no tank, block or program pickers.
Screens	Inactive tabs stay mounted; admin Today scrolls sideways; labels truncate ("D..", "M..", "Ranches & ..."); one 404 and an Animated warning on every screen.
Home	Not the prototype: account name instead of first name, no season strip, no "This week" row, tagline repeated, eight tiles, a hardcoded "Bloom Stage" and "72°F Sunny". Dashboard API returns zeros and the wrong role.
Lists	Delivery rows show the product analysis ("N-CARE 32-0-0") instead of gallons and program name; deliveries are not linked to blocks.
Tanks	Mock telemetry shown as real (42 PSI, 480 gal/week on every tank; 3,600 of 2,500 gal).
Catalogue	Still the mock list (SRavanga, Biome Care, N-CARE, K-RUSH, Bee Bloom) with free-text analysis.
Contact	A phone number on the install page and in Help; Notifications says "Push & email".
Reports	Every figure $NaN; emails instead of customer names; no month picker.
Seed data	22 near-identical deliveries, a Cancelled delivery showing "500 Gallons Delivered", test customers "Jai", john@grower.com.
Email	No sending configured; no status email observed.
Hosting	Developer's free tier at onrender.com; no custom domain; demo logins on the only build; no staging/production split.
Phase 2	Programs: empty API, no screens. Samples: submission works, review does nothing. OptimoRoute and Otodata: labels only. Install page: no device-specific steps.

3. Where it needs to be

The February 2027 release is a customer portal and an admin console that the two-person office can run without the developer: a customer sees their ranches and blocks, their delivery schedule with email updates, their invoices with the QuickBooks PDF, and a year-end summary; the office creates orders from phone calls, marks deliveries, enters invoices and runs four monthly reports. Phase 2 adds OptimoRoute delivery tracking, tank levels, nutrition programs per ranch, sample review with recommendations, and the move of hosting and accounts into ReNu-Biome's name. Prototype v2 (sections 4 and 5) shows every screen of both.

Decisions that set the scope (all made by the owner; none are open)

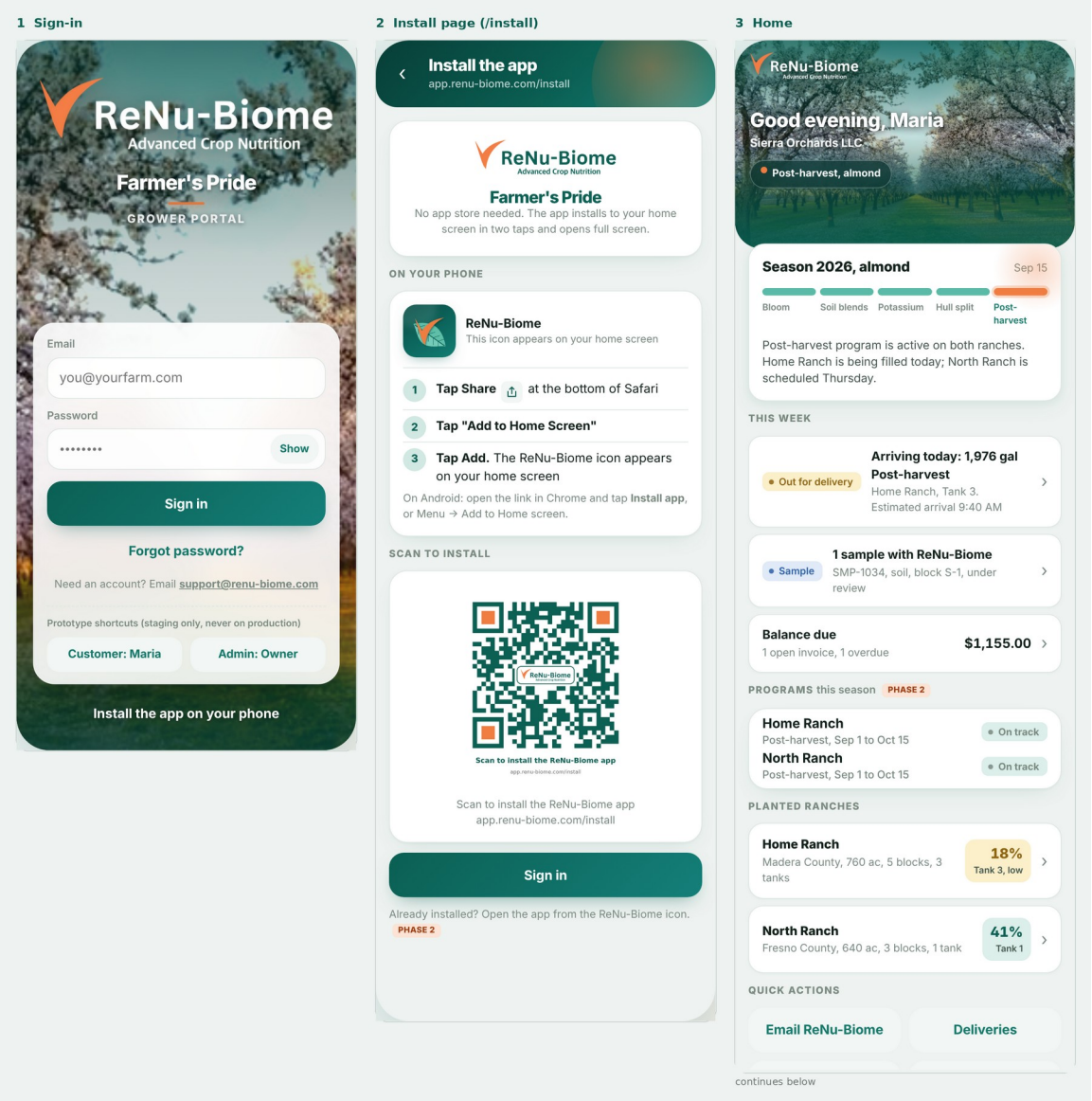
Topic	Decision	Effect on the build
Customers	One legal entity and one user per customer; every account created by the office	No entity switcher, no customer-managed users, no sign-up, no split billing
Language	English only	No Spanish, no localisation code
Products	The blend list in section 9.1 is the catalogue; sold in gallons; no prices anywhere in the app	Catalogue with analysis fields, no price fields, no pricing admin
Ordering	Customers do not order through the app; the office enters orders from phone calls and emails	Shop, cart, request, reorder, cancel stay behind the ordering flag
Delivery	Into the customer's tank; routed with OptimoRoute (Phase 2)	Delivery schedule and status in the app; OptimoRoute integration is B2
Status updates	Email only in February; OptimoRoute SMS tracking added in Phase 2 (both channels)	Email sender from support@renu-biome.com; admin setting for email / SMS / both later
Accounting	QuickBooks Online, kept by the CEO; no connection	Invoices typed in by the office with the QuickBooks PDF; QBO ID fields stored empty
Payment	On account after delivery, by check	No payments provider; in-app payment behind a flag
Statements	Year-end purchase summary per customer	Built; no monthly statements or reminders
Reports	Four monthly reports: sales by customer, sales by product, receivables aging, gallons by month; CSV and PDF	Admin More → Reports
Staff	Owner plus one office person	Two admin logins; employee role and tasks behind the staffTasks flag
Contact	Customers email support@renu-biome.com; no phone number and no hours anywhere in the app	Email button and Help text; no phone field, no response-time promise
Weather	Skipped for now	No weather chip; keep ranch

Topic	Decision	Effect on the build
		location fields for later
Map	None	The photo map is removed, not replaced
Tank levels	25% reorder threshold by default; alert goes to the office and the customer	B3; readings entered by the office first, Otodata API later
Hosting	Render, owned and paid by ReNu-Biome, pre-approved up to $40/month; app at app.renu-biome.com, API at api.renu-biome.com	B1
Pilot	The almond grower from the owner's program workbook (about 1,400 acres); the CEO as second admin; placeholders for pistachio and grape templates	B7 and section 9
Launch	February 2027, before bloom; fixed price; ReNu-Biome owns the app afterwards	Handover is part of the price (B1.9)

Design language (B8). The logo's teal-green #0C6054 everywhere the prototype shows green, with the logo's orange #F07E42 as the single accent (current-stage dot, highlighted items, the rule under the tagline); white cards with hairline borders and large rounded corners; soft tinted buttons with one gradient primary button per screen; section labels as small uppercase tracked text; status tags with a leading dot; a frosted (translucent, blurred) bottom navigation bar in the pill style; skeleton placeholders while data loads; slide-in forward and slide-back transitions where the framework allows them cheaply; Manrope or the system font. Minimum 48 px tap targets, readable in bright sun, dynamic text size (A8.8). Every green in the app, the install page and the QR code is the same green.

4. Customer app, screen by screen

The screenshots are prototype v2 as a customer sees it with Phase 2 switched on; items tagged PHASE 2 in the prototype are Phase 2 and are hidden in the February release by their feature flags. Everything else on each screen is the February release. Under each figure: what the November 2 build is missing on that screen.

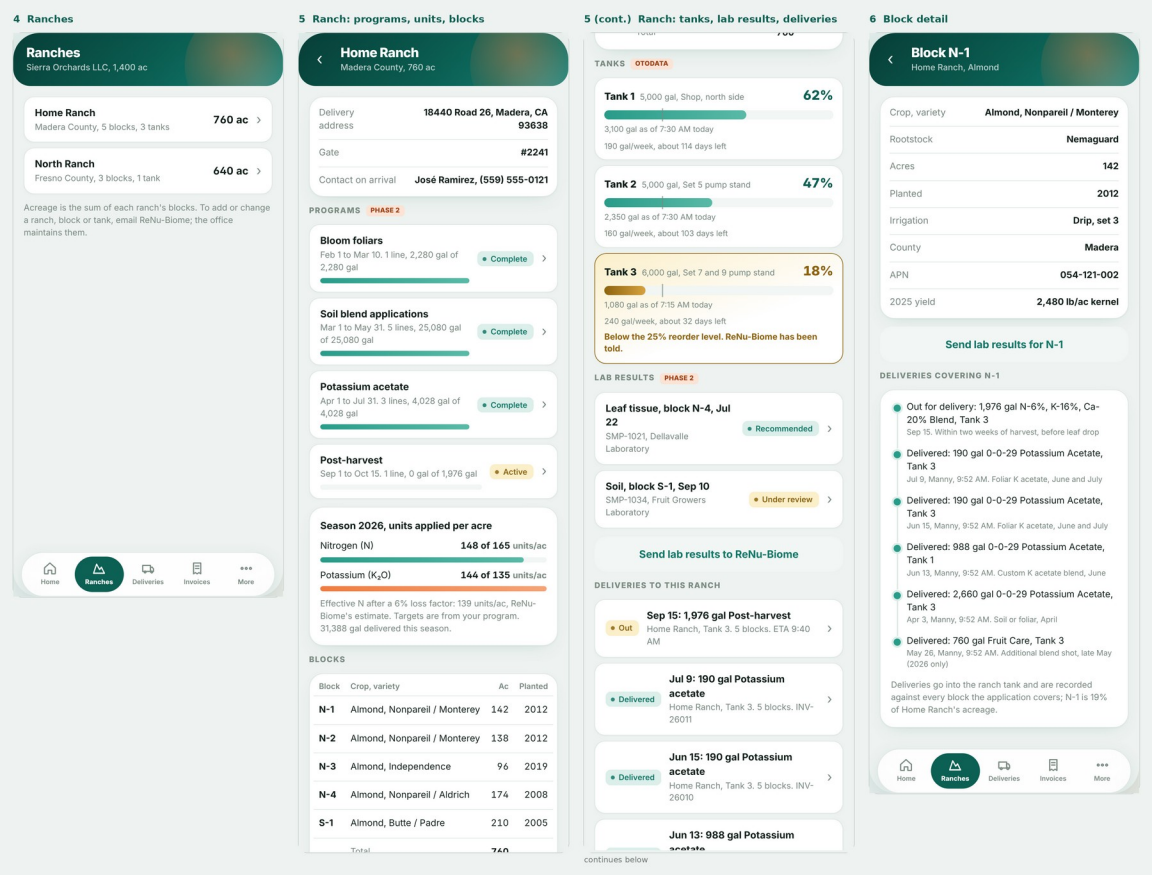


1 Sign-in (A5, B8.2). Done in the build. Keep: bloom photo under a teal gradient, logo at twice the v1 size, "Farmer's Pride" centred with the orange rule and GROWER PORTAL beneath, frosted card with Email, Password (show/hide), Sign in, Forgot password (emails a reset link), and "Need an account? Email support@renu-biome.com". Change: remove the Staff demo button now; all demo buttons disappear on production by environment variable (B1.7). Errors in plain words ("That password doesn't match"), and a wrong password keeps the email. Login from a device not seen before asks for the email code that already exists (A5.4).

2 Install page (B6). Mostly done. Change: remove the phone number; add the device-specific steps: on iPhone Safari "Tap Share, then Add to Home Screen" with the share icon drawn; on Android Chrome an "Install app" button that triggers the browser prompt, with "Menu → Add to Home screen" as the fallback; on desktop, the address and the same QR code; if already installed, go straight to sign-in. Use the light-background logo (teal wordmark, orange tick), never a tinted copy of the white one. Manifest:

name "ReNu-Biome", standalone, theme #0C6054, the 512 icon as maskable, the 180 as apple-touch-icon, the 64 as favicon (already in).

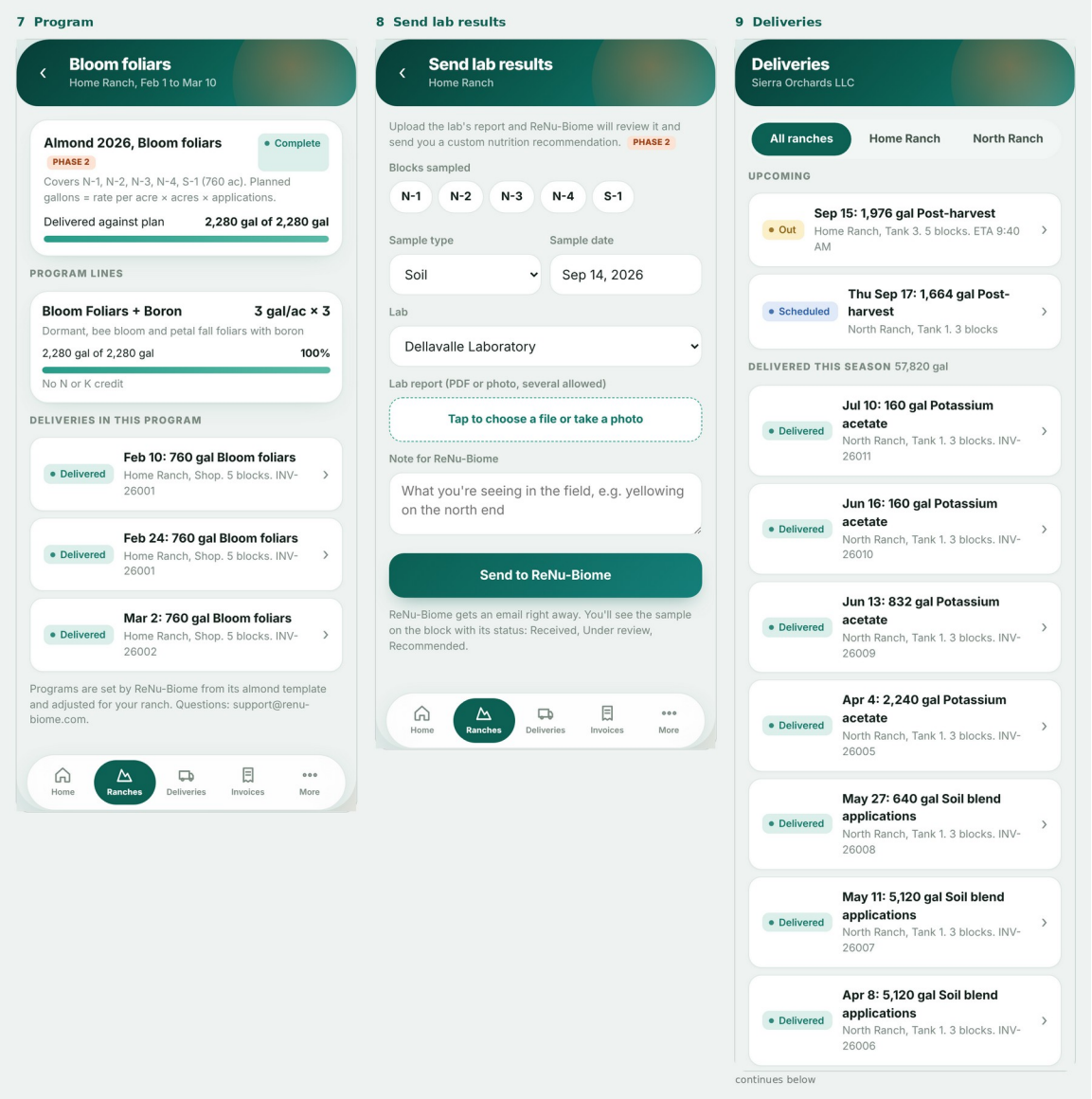
3 Home (A4, B8.3). Not done; rebuild to the figure. Photo header edge to edge with teal shade, logo top-left, a time-of-day greeting with the user's first name, the customer's name beneath, and one frosted chip with the current crop stage and the orange dot (no weather chip: deferred). The season card overlaps the header: crop and year, five-segment stage strip with the current stage highlighted, one line of status (the stage name only until programs exist). Then THIS WEEK: the next delivery (date, gallons, program name, tank, ETA when out for delivery) or "No deliveries scheduled"; a sample or recommendation waiting (flags on only); Balance due with the open-invoice count. Then PROGRAMS THIS SEASON (Phase 2) and PLANTED RANCHES cards (ranch, county, acres, blocks, tanks, and the lowest tank level once readings exist). Quick actions: Email ReNu-Biome (mailto support@renu-biome.com) and Deliveries. Remove: the account name as greeting, the repeated tagline, the eight tiles, the hardcoded "Bloom Stage" and "72°F Sunny". The dashboard API (/api/app/grower/dashboard) computes nextDelivery, openInvoiceCount, balanceDue, mainCrop and currentStage from the customer's records and returns the right role (A4.4). Bottom navigation: Home, Ranches, Deliveries, Invoices, More as a frosted pill bar.



4 Ranches (A1). Done. List of ranches with county, block and tank counts and computed acreage; the note that the office maintains ranches and blocks.

5 Ranch (A1, B3, B4, B5). Partly done. Keep the delivery address, gate and contact card (the address is new: B2 needs it). Add in order: PROGRAMS this season with status, dates, lines count and a delivered-of-planned bar (Phase 2, B4.4); the "Season, units applied per acre" card with N and K_2O against the crop targets and the effective-N line (Phase 2, section 9.3); BLOCKS as a table (block, crop and variety, acres, planted) with the total; TANKS as cards: name, capacity, location, level percent, gallons as of the reading time, weekly use and days left, with the 25% marker on the bar and an amber state below it (Phase 2, B3.4; until B3 is built, name, capacity and location only with "Level shown once a reading exists"); LAB RESULTS with status (Phase 2, B5); DELIVERIES TO THIS RANCH, newest first, with "All deliveries". Remove the mock 42 PSI and 480 gal/week values and the unrelated template text.

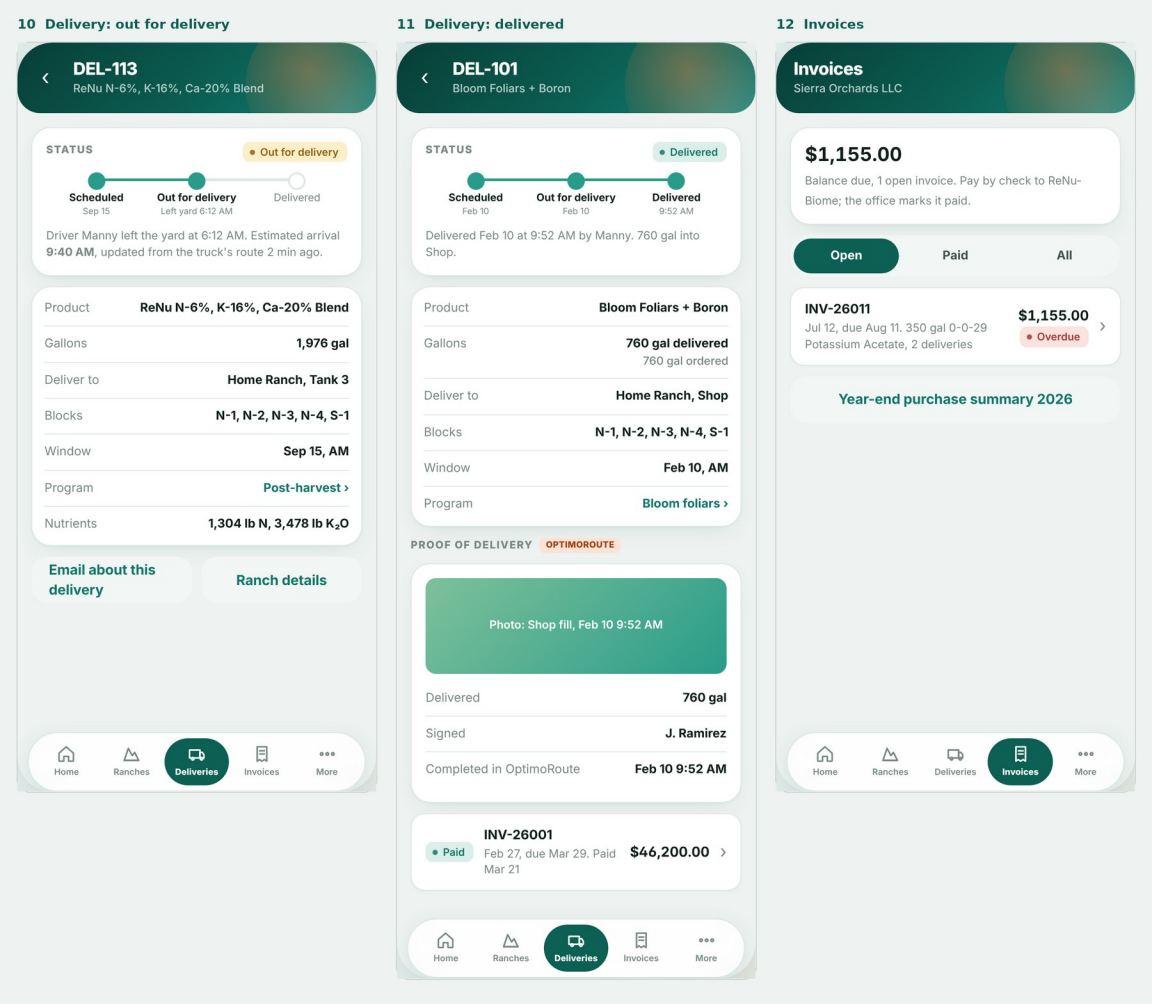
6 Block detail (A1.1). Partly done. The field card exists. Add "Send lab results for N-x" (Phase 2) and DELIVERIES COVERING N-x: a timeline of every delivery whose blocks include this one, with status, gallons, product, tank, date, driver and the program note. Deliveries must be linked to blocks (today N-4 shows no deliveries although every delivery covers all blocks).



7 Program (B4, Phase 2). Not built. Header card: crop, year, stage name, status tag, the blocks and acres covered, delivered against plan with a bar. PROGRAM LINES: product, rate per acre × applications, timing note, delivered of planned gallons with percent, and "No N or K credit" where the product carries none. DELIVERIES IN THIS PROGRAM. Footer note that programs are set by ReNu-Biome from its template.

8 Send lab results (B5.1, Phase 2). Submission exists on the build. Match the figure: blocks sampled as chips (several), sample type, sample date, lab (list plus "other"), several files (PDF or photo), note, Send. Creates a Sample with status Received and emails the office. List the sample on the ranch and the block with its status.

9 Deliveries (A2.3, B8.6). Mostly done. Change every row to the figure: status tag with dot, then "date: gallons program name" (never the blend analysis), then ranch, tank, block count, and the invoice number once invoiced or the ETA when out for delivery. UPCOMING first, then DELIVERED THIS SEASON with the season's total gallons. Ranch filter chips. Footer note: orders are placed by email with ReNu-Biome.

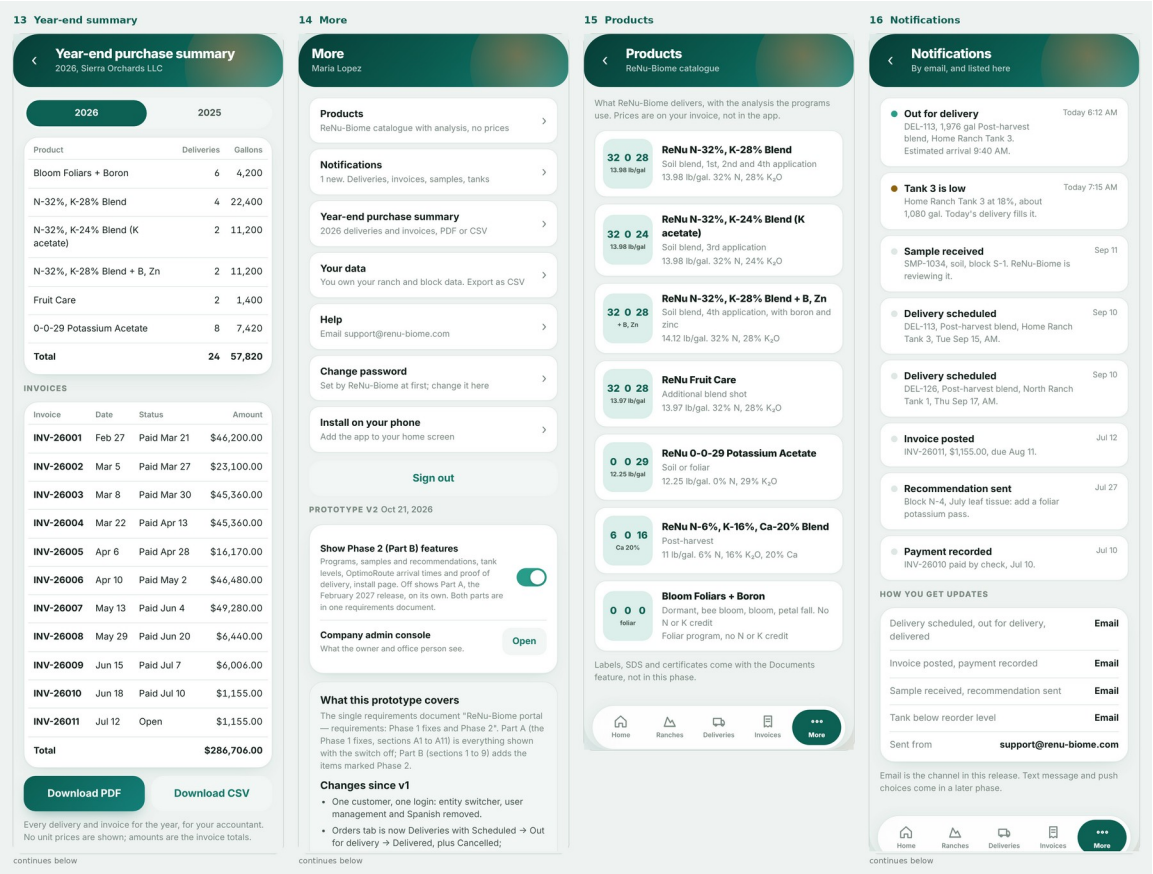


10 Delivery, out for delivery (A2.3, B2). Thin in the build. Add: the three-stage timeline with the date under each stage; the sentence with the driver, when the truck left and the estimated arrival (Phase 2,

from OptimoRoute); the fact card with product, gallons, deliver-to (ranch, tank), blocks, window, program (link, Phase 2) and nutrients in lb N and lb K_2O computed from the product analysis (section 9); buttons Email about this delivery (mailto with the delivery number in the subject) and Ranch details.

11 Delivery, delivered (A2.3, B2.4). Add: delivered gallons against ordered, the delivery time and driver, PROOF OF DELIVERY (Phase 2: photo, signature name, completion time from OptimoRoute), and the linked invoice row with its status.

12 Invoices (A3.3). Mostly done. Balance-due card with the open-invoice count and "Pay by check to ReNu-Biome; the office marks it paid"; Open / Paid / All; rows with invoice number, invoice date, due date, gallons and product, the number of deliveries, amount, status (Overdue in red past the due date). Fix: duplicate invoice numbers; show the invoice date; block names, not internal IDs. Invoice detail opens the stored QuickBooks PDF (the app-generated template only as a labelled fallback).



13 Year-end purchase summary (A3.4). Exists; match the figure: year chips, a products table (deliveries, gallons, total) and an invoices table (number, date, status with paid date, amount, total), Download PDF and Download CSV, the footer note. The PDF carries the install QR code in its footer (B6.5).

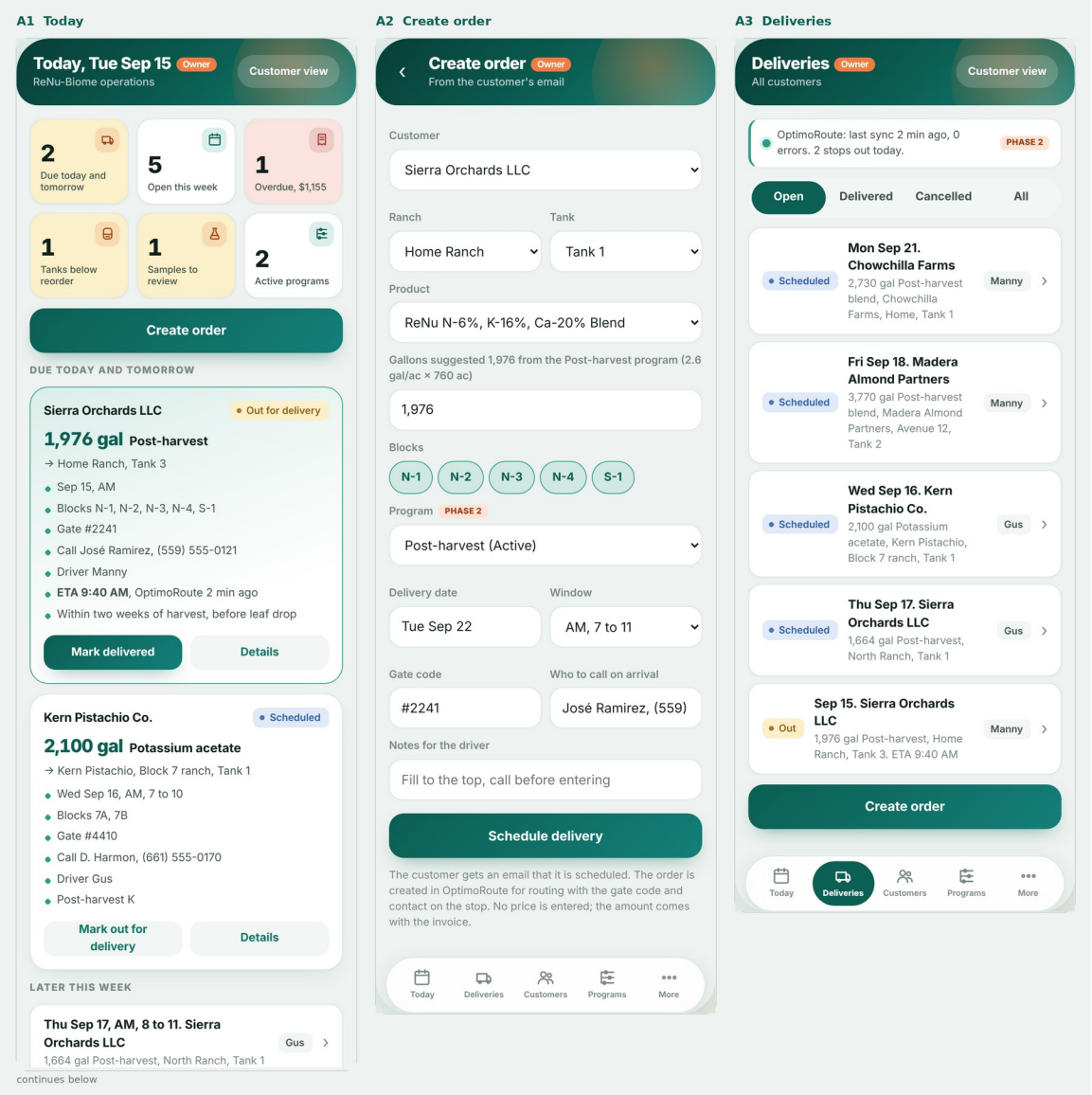
14 More (A7, A5). Keep to this list: Products, Notifications, Year-end purchase summary, Your data (CSV export of blocks and deliveries, A5.8), Help ("Email support@renu-biome.com" and what to put in the email), Change password, Install on your phone, Sign out. No phone number, no hours, no Documents, Messages, Shop or admin entry unless its flag is on. Inactive tabs are unmounted (today every tab's text is in the page).

15 Products (A6.4, section 9.1). Replace the mock catalogue. Each card: the analysis tile (N, P_2O_5, K_2O and lb/gal, or "+ B, Zn", "Ca 20%", "foliar"), name, use, and the one-line analysis. Note that prices are on the invoice, not in the app. Product images from renu-biome.com.

16 Notifications (A7.2, A7.3). Replace "Push & email alerts" with the figure: a list of system events (out for delivery with ETA, tank low, sample received, delivery scheduled, invoice posted, recommendation sent, payment recorded), then HOW YOU GET UPDATES showing Email against each event type and "Sent from support@renu-biome.com". Email is the only channel in February; the OptimoRoute SMS option is an admin setting in Phase 2 (B2.3).

5. Admin console, screen by screen

Two people use it, the owner and the office person, from a phone or a laptop (A6). Roles: Owner and Office, both admin; a Customer view button opens any customer's portal read-only with a banner (A5.5). Tabs: Today, Deliveries, Customers, Programs, More (Invoices, Products, Samples, Tanks, Program templates, Feature flags, Reports, Settings). Every green and layout rule from section 3 applies here too.

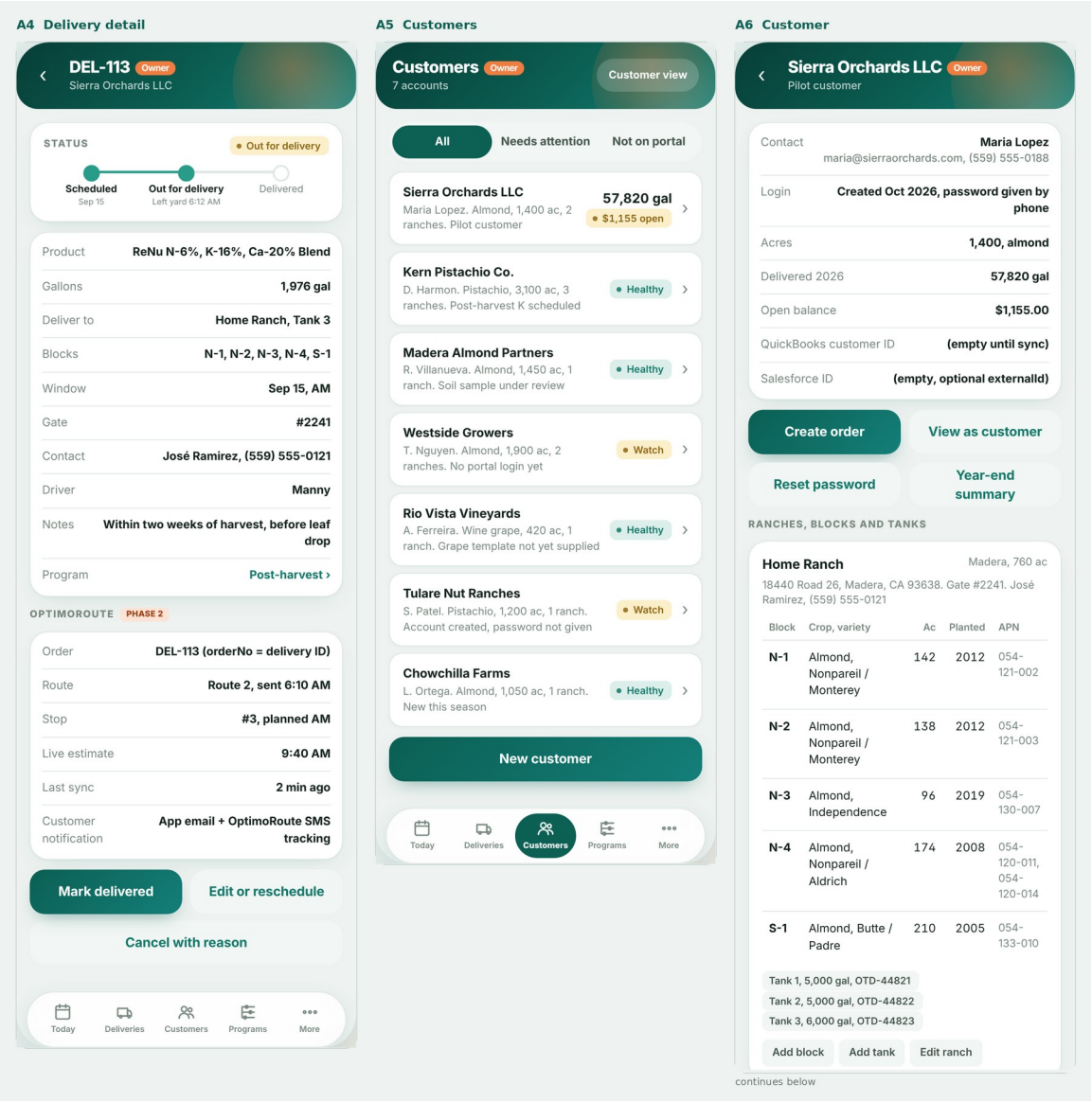


A1 Today (A6.1). Partly done. Six metric tiles with icons, computed from records: due today and tomorrow, open this week, overdue invoices with the dollar total, tanks below reorder (Phase 2), samples to review (Phase 2), active programs (Phase 2). A Create order button. DUE TODAY AND TOMORROW: one card per delivery leading with the customer name and status tag, then gallons and program name in large type, then bullets: ranch and tank, date and window, blocks, gate code, who to call with their number, driver, ETA (Phase 2) and the program note; buttons Mark delivered (or Mark out for delivery) and Details. Mark delivered opens a sheet: delivered gallons, then invoice number, invoice date, due date, amount and the QuickBooks PDF (A3.1). LATER THIS WEEK rows; NEEDS A DELIVERY (Phase 2, tanks below threshold, with Create order pre-filled); SAMPLES WAITING (Phase 2); OVERDUE INVOICES with Mark paid. Fix: customer and contact missing on cards, "Customer" placeholder on overdue rows, the sideways scroll, Create order opening the Deliveries tab.

A2 Create order (A2.1, A6.3, B4.3). Rebuild the form: Customer (from the customer list), Ranch and Tank (pickers scoped to the customer), Product (from the catalogue), Gallons with the suggestion from the

active program (rate × acres, Phase 2), Blocks as multi-select chips, Program (Phase 2), Delivery date and Window, Gate code and Who to call (pre-filled from the ranch), Notes for the driver, Schedule delivery. No price field. The footer note explains what happens: the customer gets an email; the order goes to OptimoRoute (Phase 2).

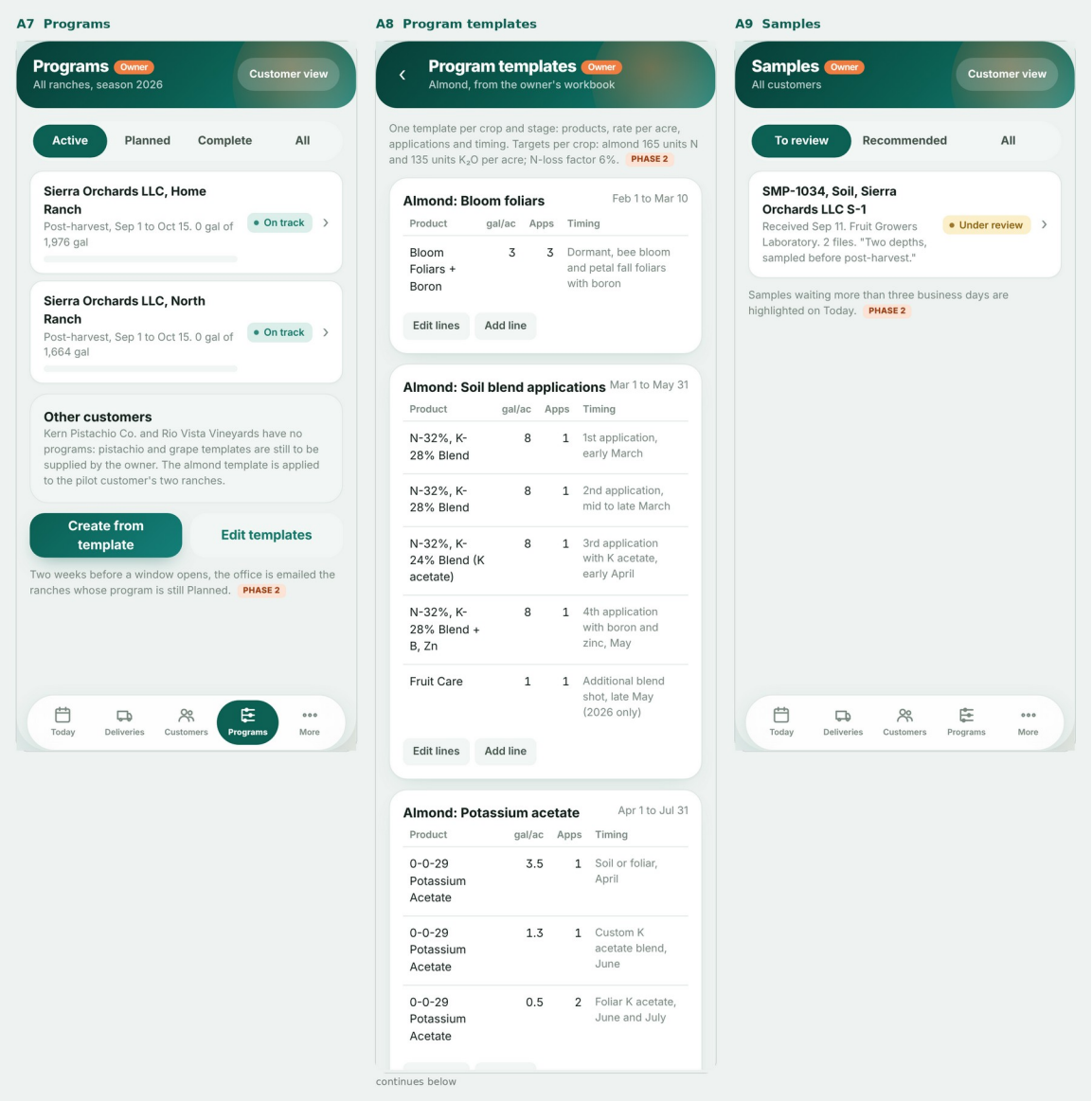
A3 Deliveries (A6.5, B2.5). Mostly done. Rows in the gallons-first style: date, customer, gallons and program name, ranch and tank, driver chip, ETA when out. Filters Open, Delivered, Cancelled, All. The OptimoRoute sync strip at the top (Phase 2: last sync, errors, stops out today).



A4 Delivery detail. Timeline; the fact card (product, gallons, deliver-to, blocks, window, gate, contact, driver, notes, program); the OPTIMOROUTE card (Phase 2: order number = delivery ID, route, stop, live estimate, last sync, customer notification channel); Mark delivered, Edit or reschedule, Cancel with reason. Cancelled by admin only; the app never overwrites a manual Cancelled with an OptimoRoute status.

A5 Customers (A6.2). The build's list exists; match the figure: name, contact, crop, acres, ranch count, a one-line status, year-to-date gallons or a health tag, open balance; filters All, Needs attention, Not on portal; New customer. Remove the test customers.

A6 Customer (A6.2, A6.2a). Contact, Login (created when, "password given by phone"), acres and crop, delivered this year, open balance, QuickBooks customer ID (empty until sync), Salesforce ID (optional externalId, section 9.6); buttons Create order, View as customer, Reset password, Year-end summary. RANCHES, BLOCKS AND TANKS: per ranch the address, gate and contact, the block table with APN, tank chips with capacity and device serial (Phase 2), Add block, Add tank, Edit ranch. Admin creates the customer's single login here: name, email, mobile, a password the admin types (A6.2a).

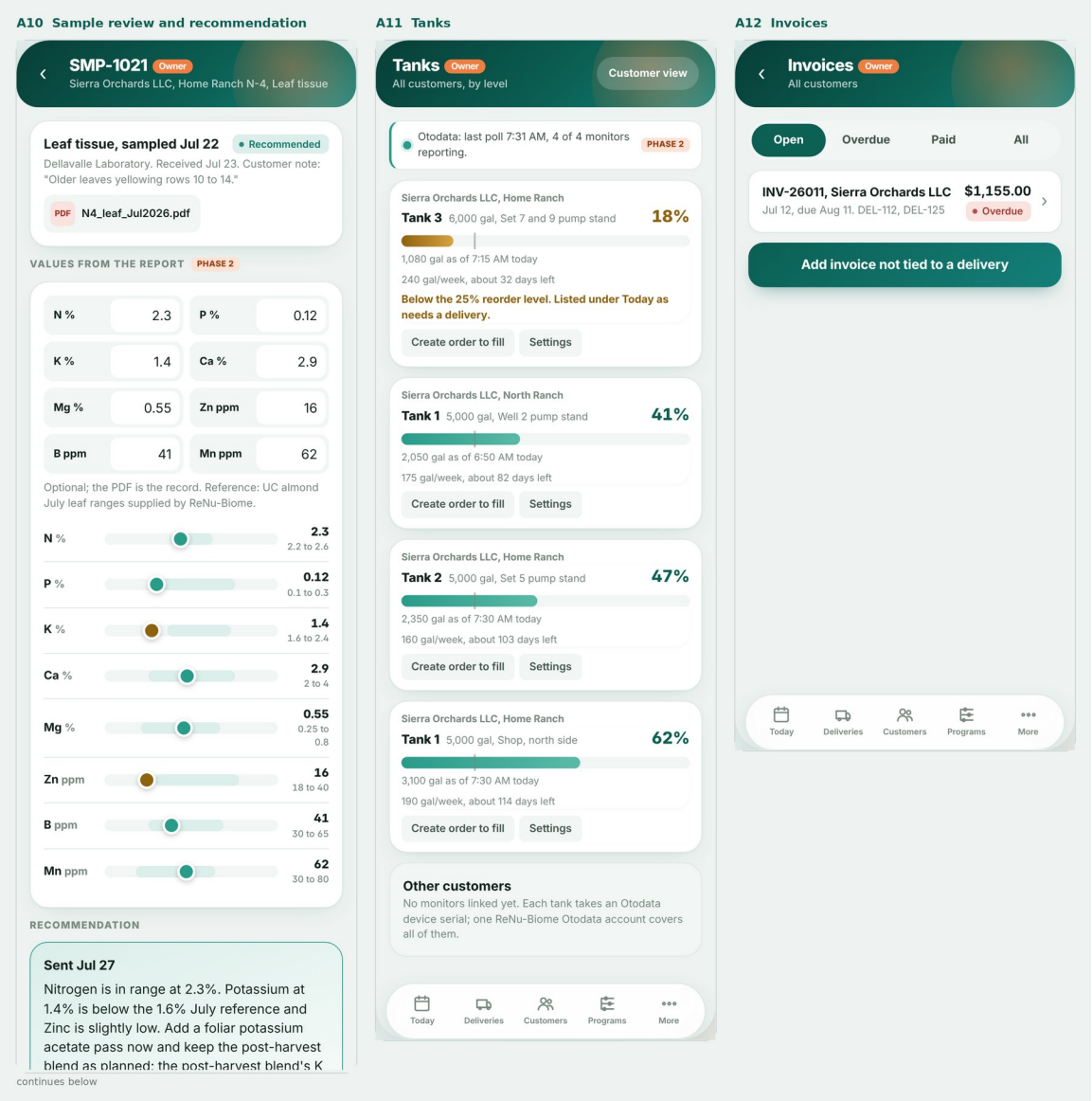


A7 Programs (B4.5, Phase 2). Not built. All ranches with their current programs: customer and ranch, stage and window, delivered of planned with a bar, On track / Behind tag (behind = Active, under 100%,

nothing scheduled); filters Active, Planned, Complete, All; Create from template, Edit templates; the note on the two-week reminder email (B4.6).

A8 Program templates (B4.1, section 9.2, Phase 2). One template per crop and stage with lines: product, gal/ac, applications, timing; Edit lines, Add line. Almond comes from the owner's workbook (section 9.2); pistachio and grape are empty placeholders with stage names only. Targets per crop (almond 165 N, 135 K_2O units/ac) and the 6% N-loss factor are fields on the template.

A9 Samples (B5, Phase 2). List with filters To review, Recommended, All; each row: sample number, type, customer and block, received date, lab, file count, customer note, status. Samples waiting more than three business days are highlighted on Today.

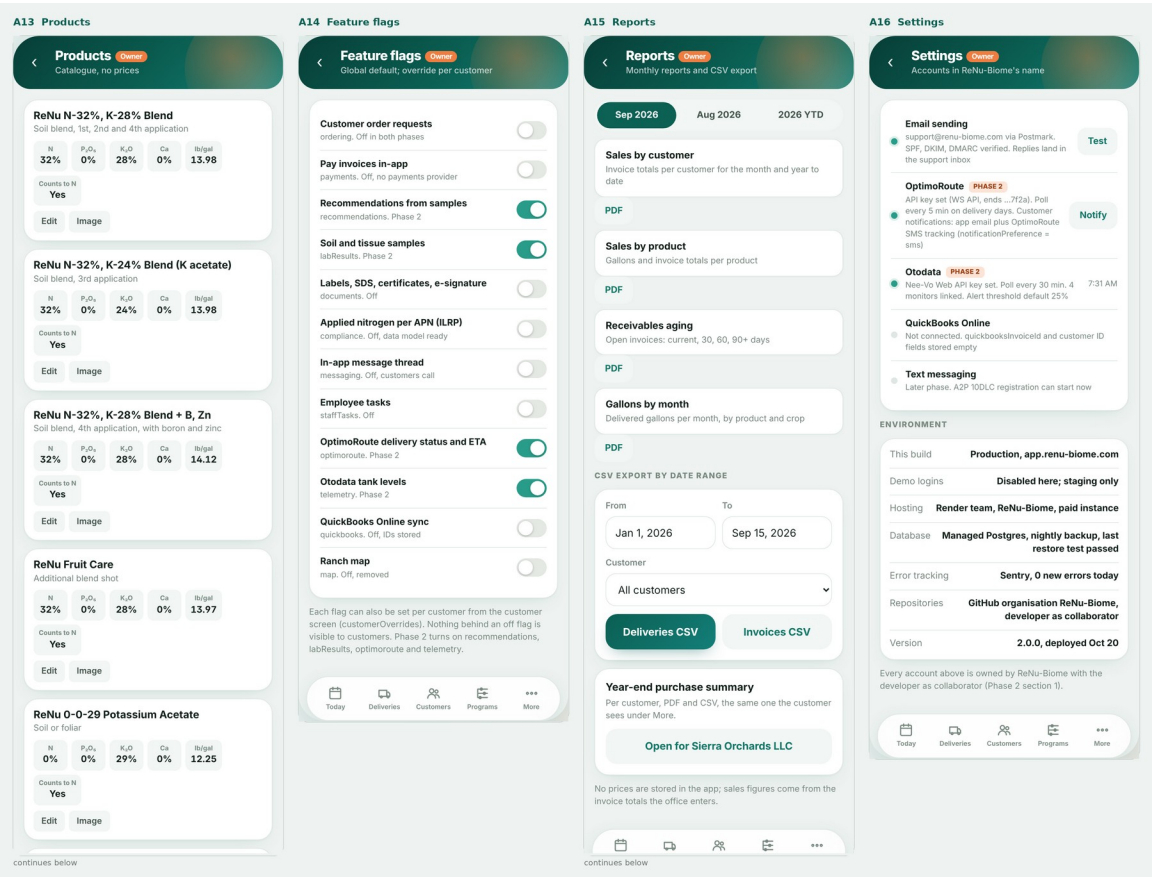


A10 Sample review and recommendation (B5.2 to B5.4, Phase 2). The build's Review Assay does nothing; build this. Header card: type, sample date, lab, received date, customer note, the attached files. VALUES FROM THE REPORT: optional numeric fields (tissue: N, P, K, Ca, Mg, S, Zn, B, Mn, Fe, Cu; soil: pH,

EC, OM, nitrate-N, P, K, Ca, Mg, Na, CEC; water: nitrate-N, EC, pH, bicarbonate), then each value drawn against the reference range for the crop and sample type with a low / in range / high dot (ranges in section 9.5, editable under Settings). RECOMMENDATION: a summary paragraph plus lines (product, rate per acre, applications, timing, blocks), Save as program or Update the active program's lines, Send. Sending sets status Recommended, emails the customer and shows the recommendation on the block and under More. Versions kept.

A11 Tanks (B3, Phase 2). All tanks across customers sorted by level: customer and ranch, tank name, capacity and location, percent, gallons as of the reading time, weekly use and days left, the 25% marker, an amber warning when below with "Listed under Today as needs a delivery"; Create order to fill (pre-filled) and Settings (threshold, device serial). Add a Record reading action per tank (level or gallons, date) for the office-entered source (B3.1). Each reading shows its source.

A12 Invoices (A6.6). Filters Open, Overdue, Paid, All; rows with number, customer, amount, invoice date, due date, linked deliveries, status; Add invoice not tied to a delivery; open a row to edit, void, mark paid, upload or replace the PDF.



A13 Products (A6.4, section 9.1). Replace the mock list. Each product: name, use line, numeric tiles for %N, $\%P_2O_5$, $\%K_2O$, %Ca and lb/gal, a Counts-to-N flag, OMRI flag (off until the owner confirms), image; Edit and Image. No price field. Delete SRavanga.

A14 Feature flags (A9). Exists. Keep: one row per flag with the global default and a per-customer override from the customer screen. The customer app must read these and hide every entry point

whose flag is off (today it ignores them). Flags: ordering, payments, recommendations, labResults, documents, compliance, messaging, staffTasks, optimoroute, telemetry, quickbooks, map.

A15 Reports (A6.7). Exists but broken ($NaN, emails instead of names, no month picker). Month chips and year to date; the four reports (sales by customer, sales by product, receivables aging current / 30 / 60 / 90+, gallons by month by product and crop) on screen with PDF; CSV export of deliveries and invoices by date range and customer; the year-end summary per customer. Sales figures are the invoice totals the office enters.

A16 Settings (B1, B2.6, B3.2, A9). Email sending: the figure's text is outdated; it must read "support@renu-biome.com via Microsoft 365" with a Test button that sends a test email to the signed-in admin. OptimoRoute (Phase 2): key set (last four characters only), poll interval, notification channel setting (email / SMS / both). Otodata (Phase 2): key set, poll interval, monitors linked, default threshold 25%. QuickBooks: not connected. Text messaging: later phase. ENVIRONMENT: build (production or staging), demo logins disabled on production, hosting, database with last backup and restore test, error tracking, repositories, version and deploy date. Reference ranges for samples are edited here (B5.3).

6. Work items, in order

Four stages: the blockers, the rest of the February release, the move to ReNu-Biome's hosting with the pilot, then the Phase 2 features. Every item can start today unless its row says otherwise; what ReNu-Biome still owes is in section 7. "Done" is what the owner will check.

Stage 1. Blockers: fix before anything else

#	Item	What to do	Done when
1	Real auth, roles and customer isolation (A5.3, A5.6)	Issue real JWTs for every login including demo accounts; enforce role and customer ID on every /api/app route; remove "mock_token"; add an automated test that logs in as two customers and confirms neither can read the other's records, including by guessing IDs, run on every deploy.	A grower token gets 403 from admin routes; the isolation test passes in CI.
2	Feature flags enforced in the UI (A9)	The client reads /api/app/feature-flags (global plus customerOverrides) and hides every entry point whose flag is off: ordering, messaging, compliance, documents, telemetry, optimoroute and the rest.	With all flags off, a customer sees only Home, Ranches, Deliveries, Invoices and the More items in screen 14.
3	Login rate limit, security headers, short token (A5.7)	Rate-limit /api/auth/login (429 after repeated failures); HSTS, CSP, X-Frame-Options, X-Content-Type-Options on API and web; access token of hours with a refresh token.	Ten wrong passwords produce a 429; headers visible in the response; the developer demonstrates at hand-over.

#	Item	What to do	Done when
4	Invoice PDF upload (A3.1, A3.3)	File field on Mark delivered and on the admin Invoices screen; store the file; View PDF opens that file; the app-generated template stays only as a labelled fallback. Any PDF serves for development; the office uploads a real QuickBooks invoice at the acceptance test.	The uploaded file opens on a phone from the customer's invoice.
5	Create order form (A2.1, A6.3)	Opens directly from Today and from a customer; customer, ranch, tank and product pickers; block multi-select; date and window; gate and contact pre-filled from the ranch; notes. Program picker comes with B4.	Screen A2 in section 5, minus the program field.
6	Unmount inactive screens; fix the overflow (A8.2)	Render only the active screen, or a navigator that unmounts; fix the sideways scroll on admin Today.	Page text of any tab contains only that tab; no horizontal scroll at 375 px.

Stage 2. The rest of the February release

#	Item	What to do	Done when
7	Home to the prototype (A4, B8.3)	Screen 3 in section 4; dashboard API computed from records (A4.4); remove the weather chip, the tagline and the tile grid.	Side by side with the prototype on a phone, recognisably the same screen with real data.
8	Delivery rows and detail (A2.3, B8.6)	Gallons and program name on every row (product short name until programs exist); detail with date, window, delivered gallons and time, invoice link, nutrients; deliveries linked to blocks and shown on the block timeline (A1.1).	Screens 9 to 11 and 6.
9	Tank cards until readings exist (A1.2)	Name, capacity and location only; "Level shown once a reading exists"; remove mock PSI, gal/week and template text.	No invented numbers on any tank.
10	Real catalogue (A6.4, 9.1)	Replace the mock list with the seven blends in section 9.1, numeric analysis fields, counts-to-N flag, OMRI flag off; delete SRavanga; product images from renu-biome.com.	Screens 15 and A13.

#	Item	What to do	Done when
11	Invoices (A3.2, A3.3)	Unique invoice numbers; invoice date on the list; block names not IDs; statuses Open, Overdue, Paid, Void; year-end summary with the QR footer.	Screens 12, 13, A12.
12	Today and Customers (A6.1, A6.2)	Customer name and who-to-call on delivery cards; names on overdue rows; the six tiles computed; customer list and customer screen as A5 and A6.	Screens A1, A5, A6.
13	Reports (A6.7)	Fix $NaN; customer and product names; month chips and year to date; PDF per report; CSV by date range and customer.	Screen A15 with correct totals on the seed data.
14	Contact wording and Notifications (A7.1, A7.2)	Remove the phone number from the install page and Help; Help text as screen 14; Notifications screen as screen 16, email only.	No phone number anywhere in the app.
15	Email sending proven (A2.4, B1.6)	Send from support@renu-biome.com through Microsoft 365 (section 7); Scheduled, Out for delivery and Delivered emails to the customer; Delivered email to the other admin; Test button in Settings.	A status email lands in Gmail, Outlook and iCloud inboxes, not spam.
16	Sign-in and install page (A5.1, B6.2)	Remove the Staff demo button; demo buttons off on production by environment variable; device-specific install steps; light-background logo.	Screens 1 and 2.
17	Labels, console, seed data (A8.4, A8.6, A8.7)	Labels that fit 375 px; clear the 404 and the Animated warning; replace the seed data with a clean fictional set shaped like the prototype's (two ranches, eight blocks, four tanks, a dated season of deliveries and invoices).	No truncated labels; clean console; seed data reads like real records.
18	Part A acceptance script	Run the February tests in section 8 with the office on a call, flags off, on a phone.	All pass.

Stage 3. Hosting, install and pilot (B1, B6, B7)

#	Item	What to do	Done when
19	Repositories (B1.1)	Transfer both repos into the ReNu-Biome GitHub organisation with GitHub's transfer (history kept); developer as collaborator with write access; branch protection on main.	Both repos show the organisation as owner; the developer can push a branch.

#	Item	What to do	Done when
20	Render team (B1.2)	The developer has accepted the invitation to the "Renu Biome App" team (Pro workspace). Move both services into it and connect them to the organisation repos; API on a 0.5 CPU / 512 MB web service, the app as a Static Site if the Expo web build serves as static files, Postgres at the $6 or $19 plan depending on daily backups. Budget $40/month; anything above, ask the owner first. Build pipeline stays on Starter.	Both services run in the team, respond within two seconds from cold, deploy from main.
21	Database (B1.3)	Managed Postgres with daily backups; migrate the current data; one restore test.	A deploy does not reset data; yesterday's backup restores to a test instance.
22	Secrets (B1.4)	Every key, JWT secret, database URL and email credential as a Render environment variable in the team; rotate the old values; a list of every secret, what it is for, where it is set.	The old JWT secret no longer authenticates; the list exists.
23	Domain and DNS (B1.5)	Custom domains app.renu-biome.com and api.renu-biome.com on the Render services; send the owner the three records (CNAME app, CNAME api, Render's verification record) for the Microsoft 365 admin to paste in; confirm auto-renewing certificates.	The app loads at app.renu-biome.com over HTTPS.
24	Environments (B1.7)	Staging (demo logins, test data) and production (demo logins off, real data only).	Production sign-in shows no demo buttons; staging does.
25	Monitoring and backups (B1.8)	Error tracking on both services in ReNu-Biome's account; health check and uptime monitor emailing the owner.	The owner receives a test alert.
26	Handover package (B1.9)	README per repo: local setup, deploy, where each secret lives, restore a backup, add a customer, the third-party accounts and owners; a runbook for deploys, restores and key rotation.	A developer who has never seen the project runs it locally from the README.
27	Cut-over (B1.10)	One agreed window outside delivery hours: deploy, repoint DNS, verify, keep the old services seven	The acceptance tests under Hosting

#	Item	What to do	Done when
		days, then delete.	in section 8 pass.
28	Production with no demo content (B7.1)	Mock customers, deliveries, invoices and the test product removed from production; real catalogue loaded.	A fresh production database holds only ReNu-Biome's real records.
29	Two real accounts (B7.2)	The office creates the CEO as admin and the pilot customer with their single login from the admin console; enters their ranches, blocks and tanks; loads the 2026 season as delivered history (dates and products in section 9.4). The developer supports; the data comes from the office after the cut-over.	Both sign in on production without help; the pilot's Deliveries tab shows the 2026 applications.
30	Install from the QR code (B7.3)	The CEO installs on an iPhone, the customer on their phone.	Both phones show the icon and open full screen.
31	One real cycle (B7.4, B7.5)	A real delivery Scheduled → Out for delivery → Delivered with emails, the invoice with its QuickBooks PDF, later marked Paid. Support path: the customer emails support@; the office emails the developer; agreed response time for the pilot weeks; the developer watches the error tracker daily for two weeks.	The February script passes on production with real data.

Stage 4. Phase 2 features (built against production, each behind its flag until accepted)

#	Item	What to do	Done when
32	Programs per ranch (B4, section 9.2 to 9.4)	Templates per crop and stage; programs created from a template onto a ranch; lines with rate × acres × applications; deliveries carry a program; progress bars in gallons and units per acre; Programs tab, Program templates, the program picker on Create order; Home and ranch sections; the two-week reminder email; pistachio and grape as empty placeholders.	Screens 5, 7, A7, A8 and the Programs tests in section 8.
33	Samples and recommendations (B5)	Submission as screen 8 (several files, blocks as chips); review screen A10 with values, reference ranges and flags; recommendation composer with Save as program; status Received → Under review → Recommended; emails both ways; history per	The Samples tests in section 8.

#	Item	What to do	Done when
		block; three-day highlight on Today.	
34	OptimoRoute (B2)	Push each delivery as an order (orderNo = delivery ID, date, type D, ranch address or tank coordinates, time window, duration, load1 = gallons, custom fields for product, tank, gate, contact); poll every five minutes on delivery days for planned time, live estimate and completion; map onto Scheduled / Out for delivery / Delivered with ETA, driver, completion time, proof of delivery; failed stops raise an admin alert; notificationPreference = sms with the customer's mobile so OptimoRoute sends its SMS tracking, with an admin setting email / SMS / both; the sync strip and the OptimoRoute card in admin. API key as OPTIMOROUTE_API_KEY in Render (the owner holds it; plan is Pro, which includes proof of delivery and realtime order tracking). Confirm Get Order Completion, Update Order Completion, Get Mobile Events and Search Orders against the live documentation before estimating; no webhooks, so poll. Needs a delivery address per ranch (the office enters it).	The OptimoRoute tests in section 8.
35	Tank levels, office-entered first (B3.1, B3.4 to B3.7)	One TankReading record (tank, level %, gallons, time, source, who); Record reading on the admin Tanks screen and on each tank; customer and admin tank cards from readings; 25% threshold per tank with one email to office and customer and a "needs a delivery" entry on Today with Create order pre-filled; weekly use and days-to-empty from the last 14 days; optional CSV importer if the Nee-Vo portal exports. No external dependency.	The first Otodata test in section 8.
36	Otodata adapter (B3.2, B3.3)	When Otodata's Nee-Vo Web API documentation arrives: device serial per tank, OTODATA_API_KEY in Render, a job every 30 minutes writing the same TankReading records with source = otodata; "no recent reading" after 24 hours silent; only the documented Web API, never the Nee-Vo app's private backend.	The remaining Otodata tests in section 8.
37	Nutrient accounting and compliance data (9.3, 9.4)	lb N and lb K_2O per delivery from the product analysis; units per acre per ranch and season against targets; effective N with the 6% loss factor; the	The units card on screen 5 matches a hand

#	Item	What to do	Done when
		compliance flag's A/R calculation defined but off.	calculation for the seed data.
38	Spreadsheet import of blocks (A11)	Import on Ranches: CSV or XLSX, one row per block, creates ranches and blocks and reports unreadable rows; Export in the same layout.	A sample file of eight blocks imports with one bad row reported.

7. Accounts, hosting, email and integrations

Every account the app depends on is owned by ReNu-Biome; the developer works as a collaborator. Secrets go into Render environment variables, never into GitHub, config files or this document.

Account	Owner	State on November 2	What the developer does
GitHub organisation	Owner (rgundavelli@gmail.com), 2FA required	Created; developer invited as collaborator	Accept; transfer both repositories (B1.1)
Render team "Renu Biome App"	Owner; company payment method; Pro workspace; $40/month spend alert	Created; developer's invitation accepted; cost pre-approved to $40/month	Move the services in; create the Postgres; set the environment variables; keep the build pipeline on Starter
Domain renu-biome.com	ReNu-Biome; DNS managed in the Microsoft 365 admin center (nameservers ns1–4.bdm.microsoftonline.com; Shopify holds only the www CNAME)	app and api subdomains decided	Send the owner three records: CNAME app → Render web service, CNAME api → Render backend, Render's verification record. The CEO pastes them in. No DNS role for the developer
Email: support@renu-biome.com (Microsoft 365 mailbox)	ReNu-Biome; the office reads the inbox	Authenticated SMTP enabled on the mailbox November 2; the developer has the mailbox sign-in	Enter SMTP_HOST=smtp.office365.com, SMTP_PORT=587, SMTP_USER and SMTP_FROM=support@renu-biome.com, SMTP_PASSWORD in Render. See the note below
OptimoRoute	CEO (admin); Pro plan: Web Service API, proof	Key received November 2 and held	Read it from OPTIMOROUTE_API_KEY in

Account	Owner	State on November 2	What the developer does
	of delivery, realtime order tracking	by the owner	Render; the owner enters it once the backend service exists
Otodata (Nee-Vo)	CEO (admin)	Documentation requested from support@otodata.ca; key and device serials to come from the CEO	Build B3 with office-entered readings first; adapter when the documentation arrives; read OTODATA_API_KEY from Render
QuickBooks Online	CEO	Not connected; no connection in scope	Store quickbooksInvoiceId and customer ID fields empty
Error tracking, uptime monitor	ReNu-Biome account	To create (B1.8)	Create in ReNu-Biome's name; alerts to the owner
Weather	none	Skipped	Keep ranch location fields; no chip

Email sending, the detail. No third-party provider and no SPF, DKIM or DMARC changes: the domain's SPF allows only Outlook senders, so sending through Microsoft 365 as the real mailbox is the only route that needs no DNS. App passwords are not offered on this tenant (Security Defaults or Conditional Access), so: (1) test a send over SMTP AUTH with the mailbox password; (2) if it is refused (535 5.7.139, or SmtpClientAuthentication disabled), set up the alternative and tell the owner exactly what the Microsoft 365 admin must do. The recommended alternative is a Microsoft Graph app registration: the admin registers an app in Entra, grants Mail.Send (application permission) with admin consent, creates a client secret, restricts the app to the support@ mailbox with an application access policy, and the tenant ID, client ID and secret go into Render as environment variables. Microsoft turns basic SMTP AUTH off by default for existing tenants at the end of December 2026 and retires it later, so build the sender so the two methods are swappable. Replies to any email land in the support inbox. The owner changes the mailbox password before the pilot goes live and updates SMTP_PASSWORD at the same time.

What ReNu-Biome has supplied

- renu-biome-brand-assets.zip: logo (dark and light backgrounds), app icon 1024/512/180/64, install QR code (PNG 300 dpi and SVG) encoding https://app.renu-biome.com/install, the stock bloom photo (placeholder, not licensed). Brand colours teal #0C6054 and orange #F07E42.
- grower-portal-prototype-v2.html, the visual reference for every screen.
- The product list with analyses, the almond program template, targets, the N-loss factor, the N-removal factor and A/R range, the pilot's 2026 application dates: all in section 9 of this document. The owner's workbook itself is not supplied.
- Reference ranges for tissue and soil values (section 9.5).
- The OptimoRoute key and plan; the support@ mailbox sign-in; GitHub and Render access.

What ReNu-Biome still owes, and when

Item	From	When
Three DNS records pasted into Microsoft 365	CEO, on the owner's request	As soon as the developer sends them (item 23)
OPTIMOROUTE_API_KEY typed into the backend service in Render	Owner	Once the service exists in the team
A delivery address per ranch; the customer's mobile number for SMS tracking	Office, in the admin console	Pilot customer first, after the cut-over
The pilot customer's ranches, blocks (with APN and 2025 kernel yield) and tanks; the pilot's contact name, email and mobile; the CEO's admin details	Office, typed into production	After the cut-over (item 29)
Otodata documentation, API key and device serials	Owner (request sent), CEO (key, serials)	Not blocking; item 36 waits on it
OMRI status per product	Owner	Later; OMRI flag stays off
Pistachio and grape templates and targets	CEO	Later; built as placeholders
The labs customers use and a sample report from each	Office	Later; build the form to the prototype's fields
A licensed bloom photo for sign-in and Home	Owner	Before production
Named owner for each account above	Owner	With this document
Agreed support response time for the pilot weeks	Owner and developer	Before the pilot customer is onboarded

What is deliberately not in scope: Spanish, a ranch map, customer ordering or payment in the app, the QuickBooks connection, in-app messaging, employee tasks, documents and e-signature, text messaging from the app (A2P 10DLC), app-store listings, Salesforce import, weather, a sample invoice before the acceptance test.

8. Acceptance tests

Run on a phone, by the office with the developer on a call. The February tests run on staging with flags off, then again on production with real data (item 31). Phase 2 tests run on staging as each feature is finished, then on production. These tests are the definition of done for the fixed price.

February release

- ☐ Admin creates a customer and its one user from the admin console and gives the customer the password by phone; the customer signs in on a page that matches screen 1; a wrong

password shows a plain message and keeps the email; a sign-in from a second device asks for the email code.

- ☐ Admin enters two ranches and six blocks by hand with crop, variety, acreage and APN; the customer sees them under Ranches with acreage computed per ranch; no map screen exists.
- ☐ Admin creates an order from a phone call (product, gallons, tank, blocks, window, gate code, contact); the customer receives an email from support@renu-biome.com and sees it under Deliveries as Scheduled with gallons and program (product) name.
- ☐ Admin marks it Out for delivery, then Delivered with the delivered gallons; the customer gets an email at each step; the delivery shows the delivered quantity and time; the other admin gets the Delivered email.
- ☐ The office enters the invoice number, dates and amount and uploads the QuickBooks PDF; it appears in the customer's Invoices linked to the delivery with gallons and product; the PDF opens on the phone.
- ☐ The office marks the invoice Paid with a date; the customer sees Paid and the balance updates; an invoice past its due date shows Overdue.
- ☐ The customer downloads a year-end purchase summary PDF and CSV listing the delivery and invoice; the PDF footer carries the install QR code.
- ☐ Home shows the photo header with greeting and first name, the customer's name, the stage chip, the season strip, the next delivery, balance due, ranch cards and the quick actions, all from real data; a new customer sees "No deliveries scheduled" and a zero balance.
- ☐ The admin Today view lists tomorrow's delivery with customer, gallons, program, ranch and tank, window, gate code and who to call as bullets, and the overdue total; Create order opens the form directly.
- ☐ The four reports show correct totals for a chosen month and year to date and download as PDF; the CSV export by date range opens in Excel.
- ☐ Shop, order request, reorder, cancel, recommendations, documents, messages, employee tasks, tank levels, OptimoRoute and the map are not visible to the customer; admin sees them under Feature flags, all off.
- ☐ A second customer's token cannot read the first customer's data (the automated isolation test is green); ten wrong passwords produce a 429; the security headers are present.
- ☐ With the phone in airplane mode, Ranches, a block, Deliveries and Invoices still display from the cache.
- ☐ No phone number anywhere in the app; no stock photos other than the agreed bloom photo; no test data; no console errors on load; no truncated labels at 375 px; every tab unmounts when left.
- ☐ Every green in the app, the install page and the QR code is #0C6054 and the accent is #F07E42.

Hosting transfer

- ☐ Both repositories are owned by the ReNu-Biome GitHub organisation; a push to main by the developer deploys to Render inside the Renu Biome App team.
- ☐ The app loads at app.renu-biome.com over HTTPS, within two seconds after an hour idle; the API answers at api.renu-biome.com.
- ☐ Production sign-in shows no demo buttons; staging does.

- ☐ A deploy does not reset data; yesterday's database backup restores to a test instance.
- ☐ The old JWT secret no longer authenticates; the secrets list matches the Render environment; no secret is in the repositories.
- ☐ A status email from support@renu-biome.com lands in Gmail, Outlook and iCloud inboxes, and a reply to it lands in the support inbox.
- ☐ The owner receives a test alert from the uptime monitor and sees the error tracker.
- ☐ A second developer, given only the README, runs the backend and frontend locally.

Install and pilot

- ☐ Production holds no demo logins, mock customers or the test product; staging still does.
- ☐ The CEO scans the QR code on an iPhone, installs from the install page in two taps, and signs in with their own admin account; the icon, name and full-screen launch are right.
- ☐ The pilot customer scans the same code, installs on their phone, signs in with the login the office created, and sees only their own ranches, blocks, tanks, deliveries and invoices, with the 2026 season as delivered history.
- ☐ A real delivery for that customer runs Scheduled → Out for delivery → Delivered with an email at each step; its invoice with PDF appears and is later marked Paid.
- ☐ The office knows the support path, and the developer confirms the error tracker shows no errors from the pilot phones after two weeks.

Programs (Phase 2)

- ☐ Admin creates a post-harvest almond program on a ranch from the template; the customer sees it on the ranch with blocks, products, rates and zero progress.
- ☐ A delivery assigned to that program moves its progress bar in gallons and in units per acre; Home shows the program as on track or behind according to the window.
- ☐ Two weeks before a Planned program's window, the office receives the reminder email.

Samples and recommendations (Phase 2)

- ☐ A customer uploads a tissue PDF for block N-4 with a note; the office receives an email and sees it as Received on Samples.
- ☐ The office enters values; potassium below the almond July leaf range shows a low flag.
- ☐ The office sends a recommendation saved as a program change; the customer gets an email, sees the recommendation and the chart on N-4, and the program lines reflect it.
- ☐ A sample older than three business days without a recommendation is highlighted on Today.

OptimoRoute (Phase 2)

- ☐ A delivery created in the app appears as an order in OptimoRoute with date, window, gate code and contact on the stop.
- ☐ After the route is sent to the driver, the app shows Scheduled with the planned arrival; when the driver starts, Out for delivery with a live estimate; when the driver completes the stop, Delivered within five minutes with the completion time, driver and proof of delivery.
- ☐ The customer receives the app's email per status change and OptimoRoute's SMS tracking messages; switching the admin setting to email only stops the SMS.

- ☐ Cancelling in the app removes the OptimoRoute order; a failed stop raises an admin alert and does not mark Delivered.

Tank levels (Phase 2)

- ☐ The office records a reading by hand on a tank; within a minute the customer's ranch screen shows that level, its time and "entered by ReNu-Biome"; a reading below the threshold produces one email to the office and the customer and a "needs a delivery" entry on Today with Create order pre-filled.
- ☐ A tank linked to an Otodata device serial shows the level and reading time the Otodata portal shows, within one reading cycle.
- ☐ A device silent for 24 hours shows "no recent reading"; the app otherwise works with Otodata unreachable.

9. Data model and rules

Customer → Ranch → Block, with tanks attached to ranches; one customer is one billing account with one login. Programs attach to a ranch and cover blocks; deliveries belong to a ranch and tank, cover blocks, and optionally serve a program; tank readings attach to tanks; samples and recommendations attach to blocks. The build already has most of this; the additions are marked.

Record	Fields	Add or change
Customer	name, contact name, email, mobile, one login, quickbooksCustomerId (empty), externalId (empty, for a later Salesforce import), notification channel setting	externalId; channel setting (Phase 2)
Ranch	name, county, delivery address, gate code, contact on arrival (name, phone), optional lat/long, acreage = sum of blocks, externalId	delivery address, lat/long, contact (needed by B2)
Block	block ID (the customer's, e.g. N-4), crop (almond, pistachio, grape), variety, rootstock, acres, planting year, irrigation type, fertigation set, county, APNs (several), kernel yield per season	in place
Tank	name, capacity gal, location, reorder threshold % (default 25), Otodata device serial (optional), optional lat/long	threshold, serial; drop the mock level and pressure
TankReading	tank, level %, gallons (level × capacity), reading time, source (manual | otodata), entered by	new (B3)
Product	name, brand label, use line, %N, $\%P_2O_5$, $\%K_2O$, %Ca, %B, %Zn, lb/gal, counts-to-N flag, OMRI flag, image, documents (flag)	numeric analysis fields; no price field anywhere

Record	Fields	Add or change
Program template	crop, stage name, stage window, lines (product, gal/ac, applications, timing note), targets per crop (N and K_2O units/ac), N-loss factor	new (B4)
Program	ranch, crop, season year, stage name, start and end, blocks covered, status (Planned, Active, Complete), lines with planned gallons (rate × covered acres × applications, overridable), notes	new (B4)
Delivery	ID (DEL-nnn), customer, ranch, tank, product, gallons ordered, blocks covered, program (optional), date, window (AM/PM or hours), gate code, contact, notes, status (Scheduled, Out for delivery, Delivered, Cancelled with reason), delivered gallons, delivered time, driver, computed lb N and lb K_2O, OptimoRoute order ID, ETA, proof of delivery (photo, signature name, completion time), historical flag	blocks link, program, nutrients, OptimoRoute fields, historical flag
Invoice	number (unique), customer, invoice date, due date, amount, status (Open, Overdue, Paid, Void), paid date, linked deliveries, PDF file, quickbooksInvoiceId (empty)	PDF file; unique number
Sample	number (SMP-nnnn), customer, blocks, type (soil, leaf tissue, petiole, irrigation water), sample date, lab, files (several), note, status (Received, Under review, Recommended), values (optional, per type), recommendation versions	review fields (B5)
Feature flag	name, global default, per-customer override	in place

9.1 Product catalogue (replaces the mock list). The website's line names are kept as a brand label on the matching blend, not as separate products. No prices.

Product	lb/gal	%N	$\%K_2O$	Counts to N	Use
ReNu N-32%, K-28% Blend	13.98	32	28	yes	Soil blend, 1st, 2nd and 4th application (2027 blends carry Ca at 15%, 10%, 20%)
ReNu N-32%, K-24% Blend (K acetate)	13.98	32	24	yes	Soil blend, 3rd application
ReNu N-32%, K-28% Blend + B, Zn	14.12	32	28	yes	Soil blend, 4th application, with boron and zinc

Product	lb/gal	%N	%K_2O	Counts to N	Use
ReNu Fruit Care	13.97	32	28	yes	Additional blend shot (2026 only)
ReNu 0-0-29 Potassium Acetate	12.25	0	29	yes	Soil or foliar
ReNu N-6%, K-16%, Ca-20% Blend	11.00	6	16	yes	Post-harvest (20% Ca)
Bloom Foliars + Boron	—	0	0	no	Dormant, bee bloom, bloom, petal fall; no N or K credit

9.2 Almond program template (from the owner's 2027 program; stage names replace the generic ones). Planned gallons for a ranch = rate × covered acres × applications.

Stage	Window	Product	gal/ac	Applications	Timing
Bloom foliars	Feb 1 to Mar 10	Bloom Foliars + Boron	3	3	Dormant, bee bloom and petal fall
Soil blend applications	Mar 1 to May 31	N-32%, K-28% Blend	8	1	1st application, early March
		N-32%, K-28% Blend	8	1	2nd application, mid to late March
		N-32%, K-24% Blend (K acetate)	8	1	3rd application, early April
		N-32%, K-28% Blend + B, Zn	8	1	4th application, May
Potassium acetate	Apr 1 to Jul 31	0-0-29 Potassium Acetate	3.5	1	Soil or foliar, April
		0-0-29 Potassium Acetate	1.3	1	Custom K acetate blend, June
		0-0-29 Potassium Acetate	0.5	2	Foliar, June and July
Post-harvest	Sep 1 to Oct 15	N-6%, K-16%, Ca-20% Blend	2.6	1	Within two weeks of harvest, before leaf drop

Targets: almond 165 units N and 135 units K_2O per acre; N-loss factor 6% for ReNu blends ("effective N", labelled as ReNu-Biome's estimate). Pistachio (stages Bloom, Nut fill, Hull split, Post-harvest) and grape (Bud break, Bloom, Veraison, Post-harvest): placeholders with stage names, no lines, no targets, until the CEO supplies them; the office fills them in from the admin console without a release.

9.3 Nutrient accounting. Per delivery: lb N = gallons × lb/gal × %N (and the same for K_2O); stored on the delivery and in the CSV export. Per ranch and season: units per acre = total lb ÷ ranch acres, shown against the crop target as the bars on the ranch screen and the program progress in units as well as gallons. Products with counts-to-N = no (bloom foliars) add nothing. Compliance (ILRP, flag off): N applied (A) from deliveries plus any irrigation-water nitrate the customer enters; N removed (R) = kernel yield × 68 lb N per 1,000 lb; A/R against the Central Valley Water Board range for mature almonds (about 1.43 to 1.44); removal factor and range per crop stored in Settings, blank for pistachio and grape; yield per block per season becomes required when the flag is on; per-APN export.

9.4 The pilot customer's 2026 season, loaded as delivered history after the cut-over (item 29), gallons = rate × block acreage once the office has entered the blocks: bloom foliars; soil blend applications on March 5, March 17 to 23, April 7 to 8 and May 9; Fruit Care on May 25 to 26; potassium acetate on April 3 to 4, June 13, June 15 and July 8 to 10. The customer is about 1,400 acres of almonds; their 2027 program is created from the almond template once B4 exists. A delivery saved as history needs no order or program: date, ranch, gallons, driver, optional product lines, flagged historical.

9.5 Reference ranges (editable under Settings; ReNu-Biome corrects them from the UC guidelines later). Almond July leaf: N 2.2 to 2.6%, P 0.10 to 0.30%, K 1.6 to 2.4%, Ca 2.0 to 4.0%, Mg 0.25 to 0.80%, Zn 18 to 40 ppm, B 30 to 65 ppm, Mn 30 to 80 ppm. Orchard soil: pH 6.0 to 7.5, EC 0 to 2.0 dS/m, organic matter 1.5 to 4%, nitrate-N 10 to 30 ppm, P 10 to 30 ppm, K 150 to 250 ppm, Ca 1,500 to 3,500 ppm, Mg 100 to 400 ppm, Na 0 to 100 ppm, CEC 10 to 25 meq. Pistachio and grape ranges: blank until supplied.

9.6 Kept empty for later. quickbooksInvoiceId and QuickBooks customer ID (sync not in scope); externalId on Customer and Ranch (Salesforce Account and Farm IDs, so a later import updates instead of duplicating); a sample's typed values are optional and the PDF is the record of truth; a delivery can be historical without an order. Nothing from Salesforce is imported in this work.

Statuses and transitions. Delivery: Scheduled → Out for delivery → Delivered; Cancelled (with reason) from Scheduled or Out for delivery, by admin only; OptimoRoute never overwrites a manual Cancelled; a failed stop stays Out for delivery and alerts admin. Invoice: Open → Paid; Open becomes Overdue past the due date; Void from any state. Sample: Received → Under review → Recommended (revisions keep the status and add a version). Program: Planned → Active → Complete; "behind" = Active, under 100% delivered, nothing scheduled. Tank alert: fires once when a reading crosses below the threshold; no repeat until a reading above it.

Emails the app sends, all from support@renu-biome.com: to the customer on Scheduled, Out for delivery (with ETA in Phase 2), Delivered (with completion time), invoice posted, payment recorded, year-end summary ready, sample received, recommendation sent, tank below reorder level; to the office on sample submitted, tank below reorder level, program reminder two weeks before a window, delivery marked Delivered by the other admin; plus the password reset and the new-device code. Every customer email carries the install QR code as a small image (B6.5).